



HK IMO Training 2021-2022
Problem Set 4
For 16 October 2021



Problem 13. Let n be a positive integer. Prove that there exists a positive integer $m > n$ such that $1 + 2^m + 3^m$ is divisible by $1 + 2^n + 3^n$.

Problem 14. The internal angle bisector of $\angle A$ of $\triangle ABC$ intersects the side BC at D and the circumcircle of $\triangle ABC$ again at E . Let M be the midpoint of BC . The circumcircle of $\triangle ADM$ intersects the sides AB and AC again at P and Q respectively. Let N be the midpoint of PQ . Let F be the foot of perpendicular from E to the line DN . Prove that the line ME is tangent to the circumcircle of $\triangle FMN$.

Problem 15. Consider an $n \times n$ chessboard with $n \geq 3$. There is a knight in the top left corner cell. In each step, the knight can move to a new cell such that

- the horizontal distance between the new cell and the old cell is 2 (where the distance of two adjacent cells is 1) and the vertical distance is 1; or
- the vertical distance between the new cell and the old cell is 2 and the horizontal distance is 1.

Three cells A, B, C form a *diagonal triple* if

- the bottom right corner of A coincides with the top left corner of B , and the bottom right corner of B coincides with the top left corner of C ; or
- the top right corner of A coincides with the bottom left corner of B , and the top right corner of B coincides with the bottom left corner of C .

Find all $n \geq 3$ such that we can choose some cells of the chessboard satisfying the following two conditions:

- it is not possible to move the knight to the bottom right corner cell without visiting any chosen cells; and
- for each diagonal triple of the chessboard, at most one cell is chosen.

Problem 16. Find the largest real constant M such that

$$\frac{x_1}{|x_2 - x_3|} + \frac{x_2}{|x_3 - x_4|} + \cdots + \frac{x_{2019}}{|x_{2020} - x_{2021}|} + \frac{x_{2020}}{|x_{2021} - x_1|} + \frac{x_{2021}}{|x_1 - x_2|} > M$$

for any pairwise distinct positive real numbers $x_1, x_2, \dots, x_{2021}$.